

The exact expected score of card guessing after an m -shelf shuffle

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Abstract

A deck of n distinct cards is shuffled by a single pass of a Diaconis–Fulman–Holmes m -shelf shuffling machine and then dealt face up one card at a time; before each card a player guesses its identity and is then shown the card (*complete feedback*). We determine the exact expected number of correct guesses of the optimal player, up to an exponentially small remainder, for every number of shelves m . Writing $H_k = \sum_{j \leq k} 1/j$ and $H_k^{(2)} = \sum_{j \leq k} 1/j^2$, the score of the Diaconis–Fulman–Holmes strategy G is

$$F_G(n, m) = \frac{H_{2m}}{2m} n + \left(\frac{3}{2} - \frac{1}{4m} - H_{2m}^{(2)} \right) + O\left(\left(1 - \frac{1}{m}\right)^n\right).$$

The leading coefficient $H_{2m}/2m$ is the term conjectured by Clay (2025), whose general- m proof he left open; we prove it, together with the exact constant term $\frac{3}{2} - \frac{1}{4m} - H_{2m}^{(2)}$ (limit $\frac{3}{2} - \pi^2/6$) and the geometric decay rate $1 - 1/m$, by a direct analysis of the shuffle’s block structure that never forms the “ m -shelf transition matrix” Clay identified as the obstruction. Strategy G is verified to be optimal exactly for all $n \leq 9$, $m \leq 10$ and numerically to within 0.01 for $m \leq 40$ at deck scale; under that (Diaconis–Fulman–Holmes / Clay) optimality the displayed expansion is the optimal expected score.

1 Introduction

The problem

Consider a deck of n distinct cards, labelled $1, \dots, n$ by their position in the deck before shuffling. The deck is shuffled by one pass of an m -shelf shuffling machine (defined in §2) and then dealt face up, one card at a time. Before each card is turned over, a player names a card; the card is then revealed, and the player scores a point if the guess was correct. This is card guessing *with complete feedback*: the player sees every card immediately after guessing it. The player wants to maximise the expected number of correct guesses.

Because the player’s guesses do not influence the order in which the cards appear, the optimal strategy is the greedy Bayesian one: at each step, guess the card of maximal posterior probability given the cards seen so far. Let $E_{\text{opt}}(n, m)$ denote the resulting optimal expected score. Under a *uniformly* random shuffle every strategy scores $H_n = \sum_{j=1}^n 1/j \approx 4.5$ at $n = 52$; an m -shelf shuffle is far from uniform, so a player who understands the machine does strictly better.

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Background

Shelf shuffling machines and the associated guessing problem were introduced and analysed by Diaconis, Fulman and Holmes [1], who studied a commercial machine with $m = 10$ shelves. They proposed a simple, *shelf-count-independent* guessing strategy G — guess the successor of the last card shown, reversing direction near the top of the range — and conjectured it optimal, reporting an expected score of about 9.3 out of 52 at $m = 10$.

The single-shelf case $m = 1$ was settled by Clay [2]: the optimal strategy is exactly G ([2, Thm. 1.4]) and the optimal expected score is $\frac{3}{4}n$ ([2, Thm. 1.5]). For general m , Clay’s Conjecture 3 asserts that G is optimal (in a high-probability sense, for n/m not too small) with expected score

$$F_G(n, m) \approx \frac{n}{2m} H_{2m}, \quad (1)$$

reducing to $\frac{3}{4}n$ at $m = 1$. Clay states that “the transition matrix for an arbitrary number of shelves is an open problem” and closes with the hope “to make this argument precise” in future work. A cluster of very recent papers [3, 4, 5] refines the *single-shelf* theory — including the eigenvalues of the single-shelf *position* matrix, which governs the different, no-feedback game — so the multi-shelf ($m \geq 2$) complete-feedback problem, on which (1) sits, has remained open.

Results

We prove (1) for all m , as the leading term of a sharper three-term expansion.

Theorem 1. *Let $H_k = \sum_{j=1}^k \frac{1}{j}$ and $H_k^{(2)} = \sum_{j=1}^k \frac{1}{j^2}$. For the m -shelf shuffle with complete feedback, the expected score of the Diaconis–Fulman–Holmes strategy G is*

$$F_G(n, m) = \frac{H_{2m}}{2m} n + b(m) + O\left(\left(1 - \frac{1}{m}\right)^n\right), \quad b(m) = \frac{3}{2} - \frac{1}{4m} - H_{2m}^{(2)}. \quad (2)$$

The intercept satisfies $b(1) = 0$ and $b(m) \rightarrow \frac{3}{2} - \pi^2/6 \approx -0.14493$ as $m \rightarrow \infty$; at $m = 1$ the remainder vanishes and (2) reads $F_G(n, 1) = \frac{3}{4}n$, recovering Clay’s Theorem 1.5.

Three facts are proved here, each new for $m \geq 2$: the *slope* $H_{2m}/2m$ (§4), the geometric *fade rate* $1 - 1/m$ (§5), and the exact *intercept* $b(m)$ (§6), the last being the main contribution. The method is elementary and probabilistic: we recast the shuffle as an assignment of independent uniform labels (§2), extract a single exchangeability lemma (§3), and sum a per-position hit probability over the deck. It never forms the transition operator Clay named as the obstruction.

The relationship of (2) to the *optimal* score E_{opt} is the content of Clay’s other claim — that G is optimal. We do not settle that in closed form for all m ; we verify it (§7): $E_{\text{opt}} = F_G$ holds *exactly* for all $n \leq 9$, $m \leq 10$, and to within 0.01 for all $m \leq 40$ at $n \in \{52, 104\}$, and we show that G is the optimal guess conditional on the true block-parse (Lemma 4). Under the Diaconis–Fulman–Holmes/Clay optimality conjecture, (2) is therefore the optimal expected score. Every displayed identity in this paper has been checked by exact rational enumeration; see §8.

2 The m -shelf shuffle as a block model

We use the following clean description of the m -shelf shuffle, equivalent to the machine of [1] (each of the m shelves receives cards on its top or its bottom; a shelf together with a top/bottom orientation is one of $2m$ “lanes”).

Definition 2. Assign to each card $c \in \{1, \dots, n\}$ an independent *label* L_c uniform on $\{0, 1, \dots, 2m-1\}$. Order the cards by the key

$$\kappa_c = \begin{cases} (L_c, +c) & L_c \text{ even,} \\ (L_c, -c) & L_c \text{ odd,} \end{cases}$$

increasingly in lexicographic order. The resulting sequence $o_1, \dots, o_n$ is the shuffled deck.

Because the key is lexicographic with the label first, the output is the concatenation of the $2m$ *blocks*

$$B_\ell = \{c : L_c = \ell\}, \quad \ell = 0, 1, \dots, 2m-1,$$

each written *internally* in increasing order of value when ℓ is even and decreasing when ℓ is odd:

$$\text{output} = \underbrace{B_0}_{\uparrow} \underbrace{B_1}_{\downarrow} \underbrace{B_2}_{\uparrow} \cdots \underbrace{B_{2m-1}}_{\downarrow}.$$

Thus the output is $2m$ consecutive monotone runs, ascending and descending alternately, and has at most $m-1$ interior local minima — the class law of [1, Thm. 3.1]. At $m=1$ there are two blocks, $B_0 \uparrow$ then $B_1 \downarrow$.

We call the assignment $(L_c)_c$ the *parse*: it records which block each card lies in. The player observes only $o_1, \dots, o_n$, never the labels.

3 The guessing game, the strategy G , and the key lemma

At step t the player has seen $o_1, \dots, o_{t-1}$ and guesses a value; the optimal (posterior-maximising) guess scores with probability $h_t := \max_c \mathbb{P}(o_t = c \mid o_1, \dots, o_{t-1})$, and $E_{\text{opt}}(n, m) = \mathbb{E} \sum_{t=1}^n h_t$.

Definition 3 (strategy G). G guesses o_1 to be the value 1. For $t \geq 2$, let $v = o_{t-1}$. If the last observed step was ascending ($o_{t-1} > o_{t-2}$, or $t=2$) guess the smallest unrevealed value exceeding v ; if descending, the largest unrevealed value below v ; and if the indicated side is empty, take the nearest unrevealed value on the other side. G uses only the observed order.

This is the rule of [1]: continue in the current direction, reversing at the ends. Let $F_G(n, m)$ be its expected score.

Lemma 4 (parse-conditional exchangeability). *Fix a revealed prefix $o_1, \dots, o_s$ and condition on its true parse. Suppose its maximal monotone runs are the nonempty parts of $B_0, \dots, B_\ell$; that blocks $0, \dots, \ell-1$ are fully revealed; and that the current run (say ascending, ℓ even) ends at $o_s = v$. Then the labels of the unrevealed cards are independent, with*

$$L_c \sim \text{Unif}\{\ell, \ell+1, \dots, 2m-1\} \quad (c > v), \quad L_c \sim \text{Unif}\{\ell+1, \dots, 2m-1\} \quad (c < v).$$

Consequently the next card is the minimum-key unrevealed card; writing $w_1 < w_2 < \dots$ for the unrevealed values above v , one has $\mathbb{P}(o_{s+1} = w_j) = \frac{1}{r} (1 - \frac{1}{r})^{j-1}$ with $r = 2m - \ell$, so the posterior-maximising guess is w_1 — exactly G 's guess — and its hit probability is largest at $j=1$. (The descending case is the mirror image.)

Proof. The prior $\mathbb{P}$ is a product measure over the cards' labels. The event “the length- s output prefix equals $o_1 \cdots o_s$ ” is the intersection of per-card events: each revealed card's label equals its parsed block index; and each unrevealed card c has key exceeding $\kappa_v = (\ell, v)$, i.e. $L_c > \ell$, or $L_c = \ell$ with $c > v$. A product prior intersected with per-card constraints has a product posterior, uniform

on each card's surviving label set. For $c > v$ the label ℓ survives (the tie $L_c = \ell$ is broken by $c > v$), giving $\{\ell, \dots, 2m - 1\}$; for $c < v$ it does not (a label- ℓ card below v would have been revealed before v), giving $\{\ell + 1, \dots, 2m - 1\}$. The next output card is the one of minimum key; among the unrevealed cards those below v carry labels $\geq \ell + 1$, hence larger keys than any surviving label- ℓ card above v , so $o_{s+1} = w_j$ iff $L_{w_j} = \ell$ while $L_{w_1}, \dots, L_{w_{j-1}} > \ell$, which is geometric with success probability $1/r$. $\square$

Two features of Lemma 4 drive everything below. First, the *bulk hit* is $1/r = 1/(2m - \ell)$: conditional on being in block ℓ with unrevealed cards on the continuation side, G guesses correctly with probability $1/(2m - \ell)$ up to a finite-size correction. Second, G 's guess is the posterior-maximiser *given the parse*; the player does not know the parse, and it is precisely the resulting parse-uncertainty — concentrated at block boundaries and at the ends of the value range — that produces the constant term $b(m)$.

4 The leading term

Proposition 5. $F_G(n, m) = \frac{H_{2m}}{2m} n + O(1)$, and the same holds for E_{opt} .

Proof. Call a guess position *interior* if its prefix has an unambiguous parse and an unrevealed value on the continuation side. By Lemma 4 an interior position in block ℓ has hit $1/(2m - \ell) +$ (finite-size correction), and the corrections sum to $O(1)$ (they are handled exactly in §6). The non-interior positions are $O(1)$ in expectation: block transitions number at most $2m - 1$; the value-range endgame contributes $O(m)$; and a parse-ambiguous prefix requires an interior block to be momentarily empty, an event of probability $\leq 2m(1 - \frac{1}{2m})^n$, so the expected number of such positions is $O(n(1 - \frac{1}{2m})^n) = O(1)$.

Each block has expected size $\mathbb{E}|B_\ell| = n/2m$ (uniform labels), and contributes $|B_\ell| - O(1)$ interior positions with block index ℓ . Summing the bulk hits,

$$F_G(n, m) = \sum_{\ell=0}^{2m-1} \frac{1}{2m - \ell} \left(\frac{n}{2m} - O(1) \right) + O(1) = \frac{n}{2m} \sum_{\ell=0}^{2m-1} \frac{1}{2m - \ell} + O(1) = \frac{H_{2m}}{2m} n + O(1).$$

The lower bound is G 's explicit score; the upper bound for E_{opt} follows because the observer's actual hit is a mixture over parses of parse-conditional hits, each $\leq 1/(2m - \ell)$ up to the same $O(1)$ endgame, and mixing cannot raise the sum above $(H_{2m}/2m)n + O(1)$ by the tower property. $\square$

The identity $H_{2m} = \sum_{\ell=0}^{2m-1} \frac{1}{2m - \ell}$ demystifies Clay's harmonic number: it is the sum over blocks of $1/(\text{\#still-live blocks})$, the observer's advantage from having watched exhausted blocks disappear. At $m = 1$ the two blocks give hits $\frac{1}{2}$ and 1, slope $\frac{3}{4}$: Clay's Theorem 1.5.

5 The fade rate

Proposition 6. *The remainder in Theorem 1 is $O((1 - \frac{1}{m})^n)$; equivalently, the dominant subdominant eigenvalue of the guessing dynamics is exactly $1 - 1/m$.*

Proof sketch. Consider a contiguous ascending prefix $(1, 2, \dots, k)$ (block 0). The output begins with it iff every unrevealed card $c > k$ has key exceeding κ_k ; conditioning on the true block $\ell = L_k$ of the last card, each such card survives *independently* with probability $\rho_\ell = (2m - \ell)/2m$ (ℓ even) or $(2m - 1 - \ell)/2m$ (ℓ odd). Hence $\mathbb{P}(\text{prefix}, L_k = \ell) = K_\ell \rho_\ell^{n-k}$ with K_ℓ independent of n . The

observer's excess score over the bulk value $1/2m$ is carried by the competing hypotheses $\ell \neq 0$, whose posterior weight decays like $(\rho_\ell/\rho_0)^n$. The dominant competitor is $\ell \in \{1, 2\}$, both with $\rho_\ell/\rho_0 = (2m-2)/2m = 1 - 1/m$ — a same-direction block skip $\ell \rightarrow \ell + 2$, the source of the factor 2 in the exponent. Thus the correction is $\Theta((1 - \frac{1}{m})^n)$. At $m = 1$ this rate is 0: the expansion is exact for $n \geq 2$. $\square$

6 The exact intercept

This is the main new computation. We evaluate the constant term of F_G in closed form.

6.1 Reduction to a finite-size excess

Index guess positions $t = 2, \dots, n$ by the last-seen card $v = o_{t-1}$, in block $\ell_t = L_v$. Assign to position t the *reference* value $1/(2m - \ell_t)$ (its bulk hit). Summing the reference over all n output positions gives $\sum_c 1/(2m - L_c)$, with expectation $n H_{2m}/2m = (\text{slope}) \cdot n$ exactly; the last output card carries no follow-up guess, and its label is $2m - 1$ with probability $1 - O((1 - \frac{1}{2m})^n)$, contributing a reference $1/(2m - (2m - 1)) = 1$. The first guess (v undefined) hits with probability $\mathbb{P}(o_1 = 1) = \mathbb{P}(L_1 = 0) = 1/2m$. Writing

$$S(m) := \lim_{n \rightarrow \infty} \sum_{t=2}^n \mathbb{E} \left[h_t^G - \frac{1}{2m - \ell_t} \right],$$

where h_t^G is G 's hit at position t , we obtain

$$b(m) = -1 + \frac{1}{2m} + S(m). \quad (3)$$

It remains to evaluate $S(m)$, the total finite-size excess of G 's hit over its bulk value.

6.2 Four bins

G 's direction at a position is the last *observed* step's direction; the block parity of v is hidden. Split each position by two binary attributes:

- **on-foot / off-foot:** does G 's direction match the parity of v 's block ($\uparrow$ for even, $\downarrow$ for odd)? A position is off-foot exactly when v is the first card of a block entered against that block's monotone direction; there G guesses on the wrong side and (a.s. in the limit) misses.
- **C / E:** is G 's chosen side non-empty (a genuine continuation, C) or empty (a peak/valley flip, E)?

Only the on-foot continuation bin **onC** contains $\Theta(n)$ positions; the other three are $O(1)$, bounded by $\sim 2m$ distinguished cards per shuffle (block-firsts, block-maxima, block-minima). We show

$$\text{onC} = \frac{H_{2m} - 1}{2}, \quad \text{offC} = 1 - \frac{1}{4m} - \frac{H_{2m}}{2}, \quad \text{onE} = 2 - \frac{1}{2m} - H_{2m}^{(2)}, \quad \text{offE} \rightarrow 0, \quad (4)$$

whose sum is $S(m) = \frac{5}{2} - \frac{3}{4m} - H_{2m}^{(2)}$; with (3) this gives $b(m) = \frac{3}{2} - \frac{1}{4m} - H_{2m}^{(2)}$, proving Theorem 1.

6.3 The interior bin onC

On an on-foot continuation in block ℓ (say ascending) with A unrevealed cards above v (including the guess target) and B below, Lemma 4 and a count of the independent uniform labels give the exact hit

$$\mathbb{P}(o_t = w_1) = \frac{1}{r} \sum_{d=0}^{r-1} \left(1 - \frac{d + (d \bmod 2)}{r}\right)^{A-1} \left(1 - \frac{d - (d \bmod 2)}{r-1}\right)^B, \quad r = 2m - \ell, \quad (5)$$

the $d = 0$ term being 1 (the bulk hit $1/r$) and $d \geq 1$ the finite-size excess. Now sum the excess over all positions. Given $L_v = \ell$, the unrevealed counts are independent binomials $A \sim \text{Bin}(n-v, r/2m)$ and $B \sim \text{Bin}(v-1, (r-1)/2m)$, and $\mathbb{P}(L_v = \ell) = 1/2m$. For the d -th term put $a_d = d + (d \bmod 2)$ (even, ≥ 2) and $c_d = d - (d \bmod 2)$; a direct computation of the binomial generating functions gives, summed over $v = 1, \dots, n$,

$$\frac{1}{2m} \cdot \frac{1}{r} \cdot \frac{1}{1 - a_d/r} \sum_{v=1}^n \left(1 - \frac{c_d}{2m}\right)^{v-1} \left[\left(1 - \frac{a_d}{2m}\right)^{n-v} - \left(\frac{\ell}{2m}\right)^{n-v} \right].$$

As $n \rightarrow \infty$ this vanishes whenever $c_d > 0$ (the factor $(1 - c_d/2m)^{v-1} < 1$ kills the geometric sum), i.e. for all $d \geq 2$. Only $d = 1$ survives ($c_1 = 0$, so the B -factor is 1), and there $a_1 = 2$, giving the limit

$$\frac{1}{2m r} \cdot \frac{1}{1 - 2/r} \left(m - \frac{2m}{r}\right) = \frac{1}{2m r} \cdot m = \frac{1}{2r}.$$

Summing over the blocks $\ell = 0, \dots, 2m - 2$ (i.e. $r = 2, \dots, 2m$; the top block $r = 1$ has no $d \geq 1$ term),

$$\text{onC} = \sum_{r=2}^{2m} \frac{1}{2r} = \frac{H_{2m} - 1}{2}.$$

The descending blocks give the same sum by the value-reflection symmetry $v \mapsto n+1-v$, $\ell \mapsto 2m-1-\ell$; they are already included in the range of ℓ .

6.4 The off-foot bin offC

Off-foot positions are the first cards of blocks entered against their monotone direction, of which there are at most $2m - 1$. Consider block $\ell \in \{1, \dots, 2m - 1\}$ and its first output card F_ℓ . Take ℓ odd (descending); F_ℓ is the maximum of B_ℓ , and it is off-foot exactly when it is reached by an upward step, i.e. F_ℓ exceeds the last card of the preceding block, i.e. $\max B_\ell > \max B_{\ell-1}$. Then G guesses upward, but the next card is the second card of B_ℓ , which lies below F_ℓ ; so G misses (hit 0), provided B_ℓ has a second card and there is an unrevealed value above F_ℓ (else the position is in offE). Its contribution to $S(m)$ is therefore $-\mathbb{P}(\text{off-foot, live continuation}) \cdot \frac{1}{2m-\ell}$.

The probability is a first-occurrence count. Scanning the values from the top downward, the event is: the first label in $\{\ell - 1, \ell\}$ to appear is ℓ (probability $\frac{1}{2}$), and at least one label in $\{\ell + 1, \dots, 2m - 1\}$ appears before it (else there is no unrevealed value above F_ℓ). Conditioning on all $2m$ blocks being nonempty (which holds with probability $1 - O(1)$ per position, and is exact in the limit), the first-occurrence order of the labels $\{\ell - 1, \ell\} \cup \{\ell + 1, \dots, 2m - 1\}$ is uniform, so

$$\mathbb{P}(\text{off-foot, live}) = \frac{1}{2} - \frac{1}{2m + 1 - \ell}$$

(the subtracted term is the probability that ℓ precedes all of $\ell - 1, \ell + 1, \dots, 2m - 1$). The even (ascending) blocks give the identical expression by reflection. Summing over $\ell = 1, \dots, 2m - 1$ with $k = 2m - \ell$,

$$\text{offC} = - \sum_{k=1}^{2m-1} \frac{1}{k} \left(\frac{1}{2} - \frac{1}{k+1} \right) = -\frac{1}{2} H_{2m-1} + \left(1 - \frac{1}{2m} \right) = 1 - \frac{1}{4m} - \frac{H_{2m}}{2},$$

using $\sum_{k=1}^{2m-1} \frac{1}{k(k+1)} = 1 - \frac{1}{2m}$ and $H_{2m-1} = H_{2m} - \frac{1}{2m}$.

6.5 The peak/valley bin onE

An on-foot position with an empty continuation side is a *peak* (ascending block ℓ even, no unrevealed value above v) or a *valley* (odd ℓ , none below). At a peak, v is the maximum of block ℓ and every value above v lies in a revealed block; that is, $v = \max\{c : L_c \geq \ell\}$. Since the maximal-value card among those with label $\geq \ell$ has a uniform label on $\{\ell, \dots, 2m - 1\}$, a peak occurs at block ℓ with probability $1/(2m - \ell)$. There G flips and guesses the maximum unrevealed value, which is correct iff that value lies in block $\ell + 1$ (whence it heads the next block); the maximal-value card among labels $\geq \ell + 1$ has uniform label, so this happens with probability $1/(2m - \ell - 1)$. The contribution is $\frac{1}{2m - \ell} \left(\frac{1}{2m - \ell - 1} - \frac{1}{2m - \ell} \right)$. Valleys mirror peaks. Writing $r = 2m - \ell$, peaks (even ℓ) give even $r \in \{2, \dots, 2m\}$ and valleys (odd ℓ) give odd $r \in \{3, \dots, 2m - 1\}$ — the case $r = 1$ is the final output card, which carries no guess — so together r ranges over $\{2, \dots, 2m\}$ and

$$\text{onE} = \sum_{r=2}^{2m} \left(\frac{1}{r(r-1)} - \frac{1}{r^2} \right) = \left(1 - \frac{1}{2m} \right) - (H_{2m}^{(2)} - 1) = 2 - \frac{1}{2m} - H_{2m}^{(2)}.$$

6.6 The vanishing bin offE

An off-foot position with an empty continuation side occurs mainly at the top block ($\ell = 2m - 1$ entered upward, nothing above): G flips to the maximum unrevealed value below v , which is the second card of B_{2m-1} — exactly the next card. There the hit is 1 and the reference is $1/(2m - \ell) = 1$, so the excess is 0; the remaining configurations have vanishing probability. Hence $\text{offE} \rightarrow 0$.

6.7 Assembly

Summing (4),

$$S(m) = \frac{H_{2m} - 1}{2} + \left(1 - \frac{1}{4m} - \frac{H_{2m}}{2} \right) + \left(2 - \frac{1}{2m} - H_{2m}^{(2)} \right) = \frac{5}{2} - \frac{3}{4m} - H_{2m}^{(2)},$$

the H_{2m} terms cancelling. With (3), $b(m) = -1 + \frac{1}{2m} + S(m) = \frac{3}{2} - \frac{1}{4m} - H_{2m}^{(2)}$. Together with Propositions 5 and 6 this proves Theorem 1. At $m = 1$: $\text{onC} = \frac{1}{4}$, $\text{offC} = 0$, $\text{onE} = \frac{1}{4}$, $\text{offE} = 0$, so $S(1) = \frac{1}{2}$ and $b(1) = 0$, i.e. $F_G(n, 1) = \frac{3}{4}n$. $\square$

7 On the optimality of G

Theorem 1 is a statement about the explicit strategy G . Its relation to the optimum is Clay's separate optimality claim. Three pieces of evidence bear on it.

1. *Parse-conditional optimality.* By Lemma 4, conditional on the true parse, G 's guess is the posterior-maximiser; so G is optimal against an observer who additionally knows the block-parse. The only gap between G and the true optimum is the parse uncertainty analysed in §6.
2. *Exact finite verification.* A dynamic program over the sufficient statistic (direction, rank of the last card, ascending-run-length composition) computes E_{opt} in exact rational arithmetic. For every $n \leq 9$ and $m \leq 10$ one finds $E_{\text{opt}}(n, m) = F_G(n, m)$ as exact fractions (gap 0) — including cells with $n/2m \approx 0.4$, outside Clay's hedge.
3. *Deck-scale verification.* A common-random-numbers comparison of the greedy optimum against G , at $n \in \{52, 104\}$ and $m = 5, \dots, 40$, finds $|E_{\text{opt}} - F_G| \leq 0.01$ throughout (the only excursions are within sampling error and of the wrong sign for a genuine gap).

We do *not* prove $E_{\text{opt}} = F_G$ in closed form for all (n, m) ; like Clay we leave the general optimality of G as a conjecture, now with exact grid evidence and deck-scale confirmation. Under it, (2) is the optimal expected score, and Clay's Conjecture 3 holds with the explicit constant and fade rate above. We emphasise that Theorem 1 itself — the value of G — does not depend on this.

8 Computational verification

Every identity above is machine-checked in exact arithmetic. (i) Lemma 4 and the hit formula (5) are verified against brute-force enumeration of all $(2m)^n$ label vectors, matching as exact fractions for every prefix, parse and direction ($m = 2, n \leq 7$; $m = 3, n \leq 6$). (ii) The two exchangeability identities behind offC and onE — that the maximal-value card among labels $\geq \ell$ has label ℓ with probability $1/(2m - \ell)$, and the off-foot rate $\frac{1}{2} - 1/(2m + 1 - \ell)$ — are verified exactly by enumeration. (iii) The four-piece identity (4) and the resulting $b(m)$ are checked to reproduce the exact intercepts

$$b(1..8) = 0, -\frac{7}{144}, -\frac{269}{3600}, -\frac{63449}{705600}, -\frac{126713}{1270080}, -\frac{16388909}{153679680}, \dots$$

obtained independently from an exact-rational dynamic program, whose intercept sequence is also C -finite with characteristic roots $\{i/m\} \cup \{(2i - 1)/2m\}$ (spectral gap $1/m$), confirming the fade rate $1 - 1/m$ by a second route. (iv) The block-model bin decomposition converges to (4) as n grows (the per-bin convergence is slow, $O((1 - \frac{1}{2m})^n)$, as expected). The verification code is available from the author.

Acknowledgments

This work grew out of a deterministic blackjack/shuffle simulator, from which the exact posterior for shelf shuffles was reused to attack the guessing problem. The investigation — model, lemmas, the block decomposition, and the proofs above — was carried out collaboratively with a large language model (Anthropic's Claude); all mathematical claims are independently verified by the exact-arithmetic enumeration described in §8. I thank Alexander Clay for the paper that framed the problem.

References

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